Can Machine Learning Reveal Heterogeneous Deterrence Effects?

A Causal-Forest Extension of Hansen (2015)

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Abstract

This report extends Hansen (2015), who uses blood alcohol content (BAC) thresholds to estimate the effect of harsher driving-under-the-influence punishments on subsequent recidivism. We first reproduce the paper's local-linear regression discontinuity design around the 0.08 BAC threshold. We then apply an honest causal forest within the same local BAC window to examine whether the estimated deterrence effect varies with driver age, gender, race, accident involvement, and year. Our conventional regression discontinuity estimate indicates that crossing the 0.08 threshold reduces four-year recidivism by approximately 2.2 percentage points, closely matching Hansen's published estimate of 2.1 percentage points. The causal-forest models produce similar average effects, but the estimated heterogeneity is generally imprecise, providing only limited evidence of systematic treatment-effect variation. The results suggest that machine learning does not materially improve the average treatment-effect estimate, but it provides an exploratory framework for examining variation that is not captured by the original homogeneous regression discontinuity model.

Keywords: regression discontinuity, causal forest, heterogeneous treatment effects, deterrence, drunk driving

1. Introduction

Economic models of deterrence predict that individuals may reduce illegal behaviour when the expected cost of punishment increases. Empirically identifying this response is difficult because individuals receiving harsher punishment often differ systematically from those receiving lighter punishment. Hansen (2015) addresses this problem using statutory BAC thresholds that generate discontinuous increases in punishment for drivers with otherwise similar measured BAC levels.

The original paper estimates an average effect at the legal threshold. However, an average effect can conceal meaningful differences across individuals. Younger and older drivers, repeat offenders, or drivers involved in accidents may respond differently to the same increase in expected punishment. Conventional subgroup regressions require the researcher to choose these interactions in advance and may become difficult to interpret when several characteristics are considered simultaneously.

This project asks:

> *Does a causal forest reveal economically meaningful heterogeneity in the effect of crossing the 0.08 BAC punishment threshold on four-year DUI recidivism?*

We first reproduce Hansen's local-linear regression discontinuity design (RDD). We then estimate separate causal forests using age, gender, race, accident involvement, and year as potential treatment-effect modifiers. We also estimate nested models that add these characteristics sequentially and repeat the analysis across several BAC bandwidths.

The machine-learning extension is evaluated according to four criteria. First, its average estimates should remain consistent with the design-based RDD benchmark. Second, estimating one characteristic at a time should produce more interpretable evidence of treatment-effect heterogeneity than placing all available variables into a single forest. Third, the findings should remain reasonably robust to the choice of bandwidth. Finally, formal difference tests are used to determine whether our estimated effects are statistically distinguishable from Hansen's published estimates.

2. Hansen's Empirical Design

Hansen (2015) studies administrative records from the Washington State Impaired Driver Program. For drivers aged 21 and older, a measured BAC of at least 0.08 constitutes a DUI, while a BAC of at least 0.15 results in enhanced penalties. Crossing these thresholds increases expected fines, licence suspensions, incarceration, and other sanctions.

The identifying assumption is that drivers immediately below and immediately above a threshold are comparable. Drivers cannot precisely control their measured BAC because it depends on consumption timing, metabolism, food intake, hydration, and measurement error. Consequently, treatment assignment near the cutoff can be interpreted as locally quasi-random, provided that potential outcomes and predetermined characteristics vary smoothly through the threshold.

Hansen estimates a local-linear RDD:

$$Y_i = \alpha + \tau D_i + \beta_1 R_i + \beta_2 (D_i \times R_i) + X_i'\gamma + \varepsilon_i, \tag{1}$$

where Y_i indicates whether driver i recidivates within four years, D_i equals one when BAC is at least 0.08, and $R_i = BAC_i - 0.08$ is the centred running variable. The interaction allows the BAC slope to differ on either side of the threshold.

Using a bandwidth of 0.05, Hansen estimates that crossing the 0.08 threshold reduces four-year recidivism by 2.1 percentage points. The corresponding estimate using a bandwidth of 0.025 is 1.9 percentage points. Both estimates are statistically significant.

3. Machine-Learning Extension

3.1. Motivation

The RDD coefficient τ in Equation (1) represents a common local treatment effect. Our extension allows the effect to vary according to predetermined characteristics:

$$\tau_i = \tau(X_i), \tag{2}$$

where X_i contains a selected driver characteristic or set of characteristics. The objective is not to replace the RDD. Instead, the causal forest is used as an exploratory method for detecting possible heterogeneous responses within the same local BAC sample.

3.2. Causal forest

We estimate an honest causal forest using the `CausalForestDML` estimator from the EconML package. Causal forests partition the covariate space into subgroups with relatively homogeneous treatment effects and estimate conditional average treatment effects by aggregating predictions across many causal trees. Honest sample splitting uses separate observations for tree construction and treatment-effect estimation, reducing bias and supporting valid statistical inference (Wager and Athey, 2018).

The outcome is four-year recidivism, and treatment equals one when BAC is at least 0.08. Centred BAC and the interaction between treatment and centred BAC are included as local RDD controls. Models use five-fold cross-fitting, 600 trees in the primary specification, and a minimum terminal-leaf size chosen to prevent very small subgroups.

The primary bandwidth is 0.05, corresponding to BAC values from 0.03 to 0.13. We separately estimate forests using:

- age;
- gender;
- race;
- accident involvement at the initial stop; and
- year.

Prior DUI experience would be an economically important treatment-effect modifier, but a consistently coded prior-DUI variable was not available in the modified dataset used for this extension. This is discussed as a limitation.

We also estimate nested models by adding age, gender, race, accident involvement, and year sequentially. Finally, we repeat the single-variable models using bandwidths of 0.015, 0.025, 0.035, 0.050, and 0.065.

Because treatment in a sharp RDD is determined by the running variable, the conventional RDD remains the primary source of causal identification. The forest results are interpreted as exploratory evidence of heterogeneity conditional on the maintained local-RDD specification.

4. Results

4.1. RDD replication

Figure 1 shows a visible decline in recidivism at the 0.08 BAC threshold. The plotted relationship is consistent with the deterrence effect reported by Hansen.

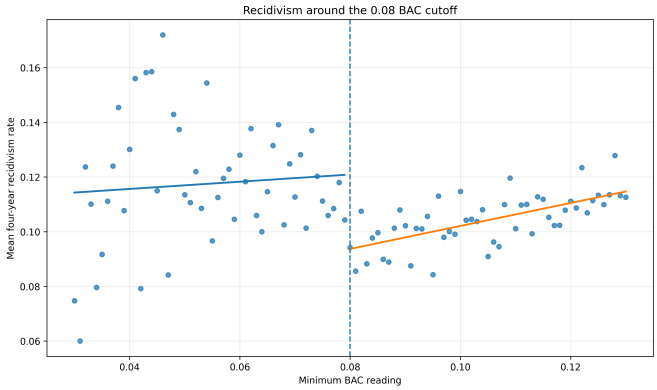


Figure 1: Recidivism around the 0.08 BAC threshold

Note: The figure presents average four-year recidivism by measured BAC and fitted local relationships on either side of the 0.08 threshold. The vertical line marks the DUI cutoff.

Table 1 reports the conventional RDD results. At the primary bandwidth of 0.05, crossing the threshold reduces recidivism by 0.022, or approximately 2.2 percentage points. The estimate is statistically significant at the 1 percent level and is close to Hansen's published estimate of 2.1 percentage points.

The estimate remains negative across all bandwidths. It ranges from a 1.3-percentage-point reduction at the narrowest bandwidth to a 2.2-percentage-point reduction at the widest bandwidth. The greater uncertainty at bandwidth 0.015 reflects its smaller sample.

Table 1: Local-linear RDD estimates

Bandwidth	Estimate	SE	95% CI	N
0.015	-0.013^{**}	0.006	$[-0.024, -0.002]$	31,294
0.025	-0.019^{***}	0.005	$[-0.029, -0.010]$	50,016
0.035	-0.019^{***}	0.004	$[-0.028, -0.010]$	67,940
0.050	-0.022^{***}	0.004	$[-0.030, -0.014]$	93,899
0.065	-0.022^{***}	0.004	$[-0.030, -0.015]$	119,295

Notes: The dependent variable is an indicator for recidivism within four years. Regressions use a rectangular kernel, side-specific local-linear BAC trends, available predetermined controls, and standard errors clustered at the 0.001 BAC level. $^{***}p < 0.01$, $^{**}p < 0.05$.

The balance tests provide no evidence of discontinuities in observed predetermined characteristics. At bandwidth 0.05, the estimated jumps in gender, race, age, and accident involvement are all statistically insignificant. These tests support, although cannot prove, the local comparability assumption underlying the RDD.

4.2. Single-characteristic forests

Table 2 reports the average effect from each single-characteristic forest. The estimates are tightly concentrated between -0.0223 and -0.0226. This similarity shows that allowing the forest to use different effect modifiers does not materially change the average deterrence estimate.

Table 2: Single-characteristic causal-forest estimates

Modifier	ATE	SE	95% CI	CATE SD
Age	-0.0225	0.0262	$[-0.0738, 0.0289]$	0.0254
Male	-0.0226	0.0053	$[-0.0329, -0.0123]$	0.0071
White	-0.0224	0.0049	$[-0.0320, -0.0127]$	0.0062
Accident	-0.0224	0.0062	$[-0.0346, -0.0103]$	0.0042
Year	-0.0223	0.0125	$[-0.0468, 0.0022]$	0.0095

Notes: Each row is obtained from a separate causal forest in which the named variable is the treatment-effect modifier. ATE denotes the average estimated treatment effect. CATE SD is the standard deviation of the individual conditional-effect predictions. All models use bandwidth 0.05 and $N = 93{,}899$.

The gender-, race-, and accident-based models produce confidence intervals that exclude zero. By contrast, the age and year models are considerably less precise. Importantly, statistical significance of the average effect within a forest does not establish that effects differ significantly between the categories of the named characteristic.

Figure 2 compares these average forest estimates with Hansen's published result. Every point estimate lies close to both the replicated RDD estimate and Hansen's estimate. Machine learning therefore does not produce a materially different estimate of the average effect.

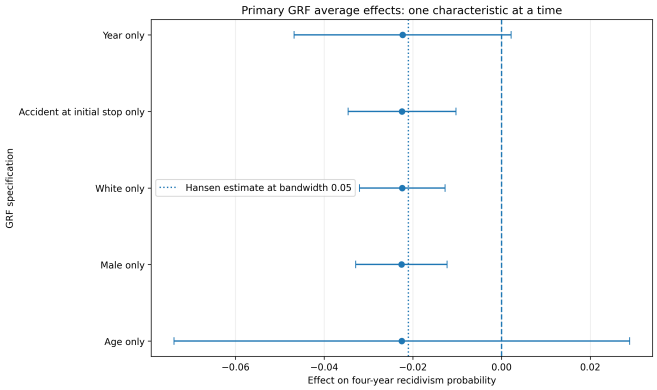


Figure 2: Average effects from single-characteristic forests

Note: Each estimate comes from a separate forest using the named variable as an effect modifier. The estimates do not represent effects only for the named subgroup.

4.3. Comparison with Hansen's Estimate

To determine whether our estimates differ statistically from Hansen's published result, we conduct an approximate difference-in-means test. The test statistic is

$$z = \frac{\widehat{\tau}_{\text{ours}} - \widehat{\tau}_{\text{Hansen}}}{\sqrt{SE(\widehat{\tau}_{\text{ours}})^2 + SE(\widehat{\tau}_{\text{Hansen}})^2}}, \tag{3}$$

where Hansen's estimate and standard error are treated as independent from our estimates. The null hypothesis is that the two estimated treatment effects are equal.

Table 3: Comparison with Hansen's Published Estimates

Estimate	Ours	Hansen	Diff.	z	p
RDD ($h = 0.050$)	-0.0222	-0.0210	-0.0012	-0.202	0.840
RDD ($h = 0.025$)	-0.0195	-0.0190	-0.0005	-0.065	0.948
Forest ($h = 0.050$)	-0.0225	-0.0210	-0.0015	-0.056	0.956

Notes: The null hypothesis is that our estimate equals Hansen's published estimate. The test statistic assumes independent estimates and uses the sum of the estimated variances. Because Hansen's data and our modified data may overlap, these tests should be interpreted as approximate comparisons.

The estimated differences from Hansen's published results are small and statistically insignificant. At bandwidth 0.05, the conventional RDD estimate is approximately 0.12 percentage points more negative than Hansen's estimate, but the equality test produces a p-value of 0.840. At bandwidth 0.025, the difference is less than 0.05 percentage points, with a p-value of 0.948. The age-only causal-forest estimate at the primary bandwidth is also statistically indistinguishable from Hansen's estimate, with a p-value of 0.956. We therefore fail to reject the null hypothesis that our estimates are equal to Hansen's published estimates.

4.4. Heterogeneity Profiles

Figure 3 presents the estimated treatment-effect profile by age. Among the characteristics considered, age produces the greatest dispersion in predicted treatment effects. The point estimates vary across age groups, with some of the most negative estimates appearing among older drivers. However, the confidence intervals are wide and overlap substantially across groups. The age profile therefore provides suggestive evidence of heterogeneity, but it does not establish that the deterrence effect differs systematically by age.

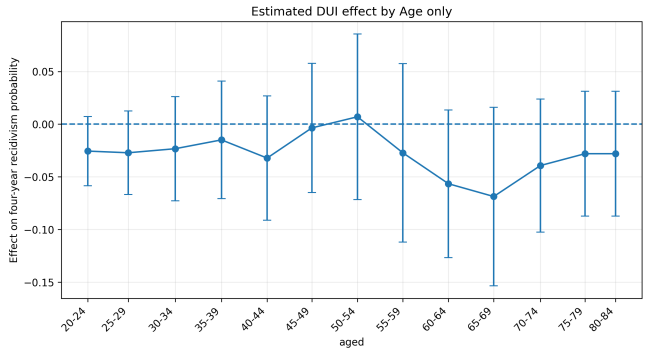


Figure 3: Estimated Treatment-Effect Profile by Age. *Note:* Points report average predicted conditional average treatment effects within five-year age groups. Vertical bars represent 95% confidence intervals. Negative estimates indicate a reduction in four-year recidivism associated with crossing the 0.08 BAC threshold.

The remaining treatment-effect profiles are reported in the appendix. The gender, race, and accident profiles show modest differences between men and women, white and non-white drivers, and accident and non-accident stops. The year profile also displays some variation over time. In each case, however, the corresponding confidence intervals overlap considerably. These patterns should therefore be interpreted as descriptive rather than as evidence of statistically significant subgroup differences.

We do not conduct formal pairwise equality tests between subgroup CATEs. Consequently, the profile figures are used to identify suggestive patterns rather than to establish that one group responds differently from another. Statistical significance of the average treatment effect within a single-characteristic forest does not, by itself, imply statistically significant treatment-effect heterogeneity across the values of that characteristic.

The limited evidence of heterogeneity has two possible interpretations. First, the deterrence effect may genuinely be relatively similar across the observed driver groups near the threshold. Second, the available characteristics may be too coarse to capture the most economically relevant sources of variation. Prior DUI experience, previous punishment exposure, income, addiction severity, and the sanctions actually imposed may be more important treatment-effect modifiers, but these variables are not available in the modified dataset.

4.5. Nested and Bandwidth Results

The nested models produce average treatment effects between -0.0225 and -0.0230, regardless of how many characteristics are included. However, precision decreases as additional effect modifiers are added. The confidence interval for the full model ranges from approximately -0.093 to 0.047, indicating considerably greater uncertainty than the corresponding single-characteristic models.

These results suggest that increasing model complexity does not improve the average treatment-effect estimate. Although the larger feature set allows the causal forest to consider a greater number of possible treatment-effect partitions, the available data do not provide sufficient information to estimate all resulting local effects precisely.

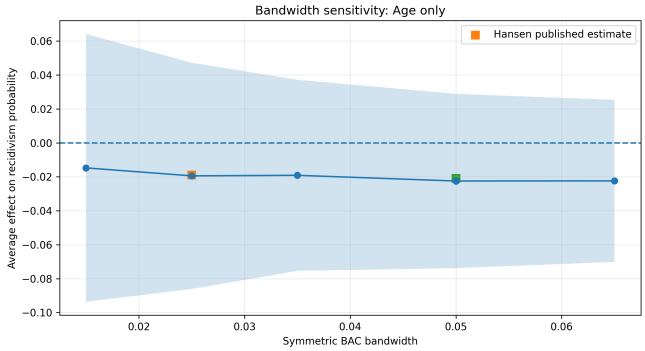


Figure 4: Bandwidth Sensitivity of the Age-Only Causal Forest. *Note:* Points report the average estimated treatment effect from the age-only causal forest using symmetric BAC bandwidths of 0.015, 0.025, 0.035, 0.050, and 0.065 around the legal BAC threshold. The shaded region represents the 95% confidence interval. The orange and green markers indicate Hansen's estimates at bandwidths 0.025 and 0.050.

Figure 4 illustrates the bandwidth sensitivity of the age-only causal forest. Across bandwidths of 0.015, 0.025, 0.035, 0.050, and 0.065, the estimated average treatment effects remain consistently negative and are concentrated around a reduction of approximately two percentage points. This stability indicates that the estimated deterrence effect is not driven by the primary bandwidth of 0.050.

The principal difference across bandwidths is precision rather than the magnitude of the estimated effect. Narrower bandwidths compare drivers who are more similar in BAC, strengthening the local interpretation of the regression discontinuity design, but they also contain fewer observations and therefore produce less precise estimates. Wider bandwidths increase the available sample size and modestly improve precision, although the 95% confidence intervals continue to include zero for the age-only model. Consequently, the age profile remains statistically imprecise despite the stability of its point estimates.

Although Figure 4 presents only the age-only model, the remaining bandwidth analyses for gender, race, accident involvement, and year exhibit similar qualitative patterns (see Appendix Figures A6–A9). Across these models, point estimates remain relatively stable as the bandwidth expands, while any gains in precision are modest. Overall, changing the bandwidth does not materially alter the estimated average treatment effect or the substantive conclusions of the analysis.

5. Economic Interpretation

The primary RDD estimate implies that drivers immediately above the 0.08 BAC threshold are approximately 2.2 percentage points less likely to recidivate within four years than otherwise comparable drivers immediately below it. Relative to a below-threshold recidivism rate of approximately 11.7 percent in our sample, this corresponds to a reduction of about 19 percent.

This effect is economically meaningful. Crossing the legal threshold exposes drivers to substantially higher expected sanctions, including larger fines, licence restrictions, possible incarceration, higher insurance costs, and more severe penalties for future offences. The observed reduction in recidivism is therefore consistent with drivers responding to the higher expected cost of violating impaired-driving laws.

The machine-learning extension does not overturn this conclusion. Instead, it shows that the estimated average treatment effect is highly stable across the observed effect modifiers. Although the causal forests produce some variation in individual treatment-effect predictions, the estimates are not sufficiently precise to conclude that deterrence differs systematically by age, gender, race, accident involvement, or year.

From a policy perspective, these findings suggest that the available data do not provide evidence to support different punishment policies for the observed driver groups. The primary policy implication therefore remains consistent with Hansen's original findings: the discrete increase in sanctions at the legal BAC threshold appears to reduce subsequent drunk driving.

6. Limitations

Several limitations affect the interpretation of the extension.

First, treatment is deterministically assigned by BAC in a sharp RDD. Identification comes from continuity at the threshold rather than from standard selection-on-observables assumptions. The causal forest is therefore best viewed as an exploratory extension built around the local RDD, not as an independent causal design.

Second, the forest can identify only heterogeneity associated with observed variables. The modified dataset does not provide a consistently coded prior DUI variable, although Hansen's findings suggest that prior offenders may respond more strongly

to punishment. Other potentially important variables, including income, employment, addiction severity, exact sanctions, and access to transportation, are also unavailable.

Third, estimating many individual and subgroup effects creates a risk of finding apparent patterns by chance. The profiles should therefore be interpreted cautiously, particularly where confidence intervals are wide or where subgroup comparisons were not prespecified.

Fourth, wider bandwidths improve precision by including more observations but move farther away from the cutoff. This creates a trade-off between statistical power and the credibility of the local-linear approximation.

Finally, the results apply to tested drivers in Washington State during the study period. They need not generalize to other jurisdictions, enforcement systems, legal thresholds, or populations.

7. Conclusion

This report extends Hansen's RDD analysis using causal forests to examine heterogeneous deterrence effects. The replicated local-linear RDD indicates that crossing the 0.08 BAC threshold reduces four-year recidivism by approximately 2.2 percentage points, closely matching Hansen's published estimate.

The causal-forest models generate nearly identical average effects across age, gender, race, accident involvement, and year. The one-characteristic models provide more interpretable profiles than the full nested model, but the confidence intervals offer limited evidence that the effects are statistically different across groups. The estimates also remain stable across alternative bandwidths, and the approximate equality tests fail to reject that our primary estimates are equal to Hansen's published results. Although the predicted individual effects display some variation, the strongest patterns are imprecisely estimated. Adding more characteristics increases uncertainty without materially changing the average estimate.

Machine learning therefore contributes primarily as an exploratory and interpretive tool in this application. It provides a flexible way to search for possible heterogeneity, but it does not improve the precision or policy relevance of the original average-effect estimate. The conventional RDD remains the clearest and most credible empirical result.

References

Hansen, B. (2015). Punishment and deterrence: Evidence from drunk driving. *American Economic Review*, 105(4):1581–1617.

Wager, S. and Athey, S. (2018). Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523):1228–1242.

A. Additional Results

A.1. Balance tests

Table 4: Continuity of predetermined characteristics

Characteristic	Estimate	SE	95% CI	p-value	Mean below cutoff
Male	0.0061	0.0051	$[-0.0039, 0.0160]$	0.232	0.788
White	0.0017	0.0055	$[-0.0092, 0.0126]$	0.760	0.847
Age	-0.1470	0.1659	$[-0.4721, 0.1780]$	0.375	34.940
Accident	-0.0008	0.0035	$[-0.0077, 0.0060]$	0.807	0.099

Notes: Each row reports a separate local-linear RDD using the predetermined characteristic as the dependent variable. All regressions use bandwidth 0.05, a rectangular kernel, and standard errors clustered at the 0.001 BAC level. The sample contains 93,899 observations and 101 BAC clusters.

A.2. Nested Causal-Forest Models

Table 5 reports the primary-bandwidth causal-forest results when characteristics are added sequentially. The point estimates remain close to a 2.3-percentage-point reduction in recidivism, but the standard errors increase as the number of effect modifiers grows.

Table 5: Nested Causal-Forest Estimates

Specification	ATE	SE	95% CI	p-value
Age	-0.0225	0.0262	$[-0.0738, 0.0289]$	0.391
Age + Male	-0.0227	0.0312	$[-0.0838, 0.0384]$	0.467
Age + Male + White	-0.0229	0.0324	$[-0.0863, 0.0406]$	0.480
Age + Male + White + Accident	-0.0230	0.0327	$[-0.0871, 0.0412]$	0.483
Full model	-0.0227	0.0357	$[-0.0927, 0.0473]$	0.525

Notes: All models use a symmetric BAC bandwidth of 0.05 and contain 93,899 observations. Characteristics are added sequentially in the order shown. The full model includes age, gender, race, accident involvement, and year. ATE denotes the average estimated treatment effect.

A.3. Full Comparison with Hansen

Table 6 expands the comparison reported in the main text to all single-characteristic causal-forest models at the primary bandwidth. None of the estimated effects is statistically distinguishable from Hansen's published estimate.

Table 6: Full Comparison with Hansen's Published Estimate

Model	Estimate	Difference	z	p-value
RDD	-0.0222	-0.0012	-0.202	0.840
Age only	-0.0225	-0.0015	-0.056	0.956
Male only	-0.0226	-0.0016	-0.237	0.812
White only	-0.0224	-0.0014	-0.218	0.828
Accident only	-0.0224	-0.0014	-0.194	0.846
Year only	-0.0223	-0.0013	-0.099	0.921

Notes: All rows use a bandwidth of 0.05 and are compared with Hansen's published estimate of -0.021 with a standard error of 0.004. Difference is our estimate minus Hansen's estimate. The tests use the sum of the estimated variances and assume independence between estimates. Because the underlying samples may overlap, the reported tests are approximate.

A.4. Effect profiles

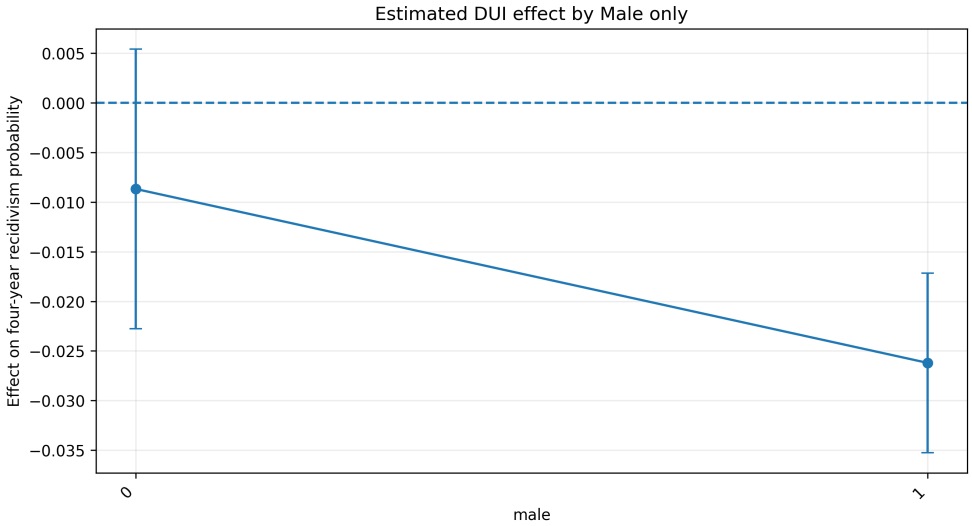


Figure 5: Estimated treatment-effect profile by gender

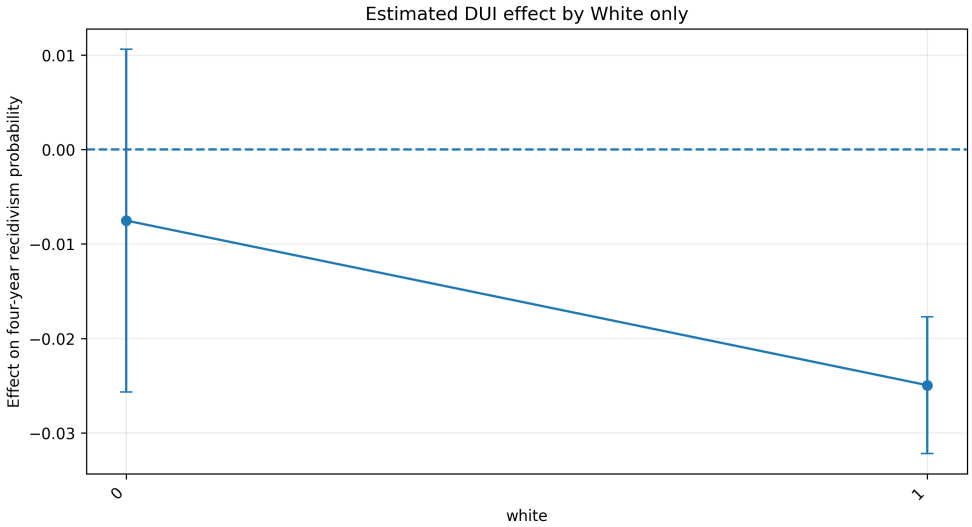


Figure 6: Estimated treatment-effect profile by race

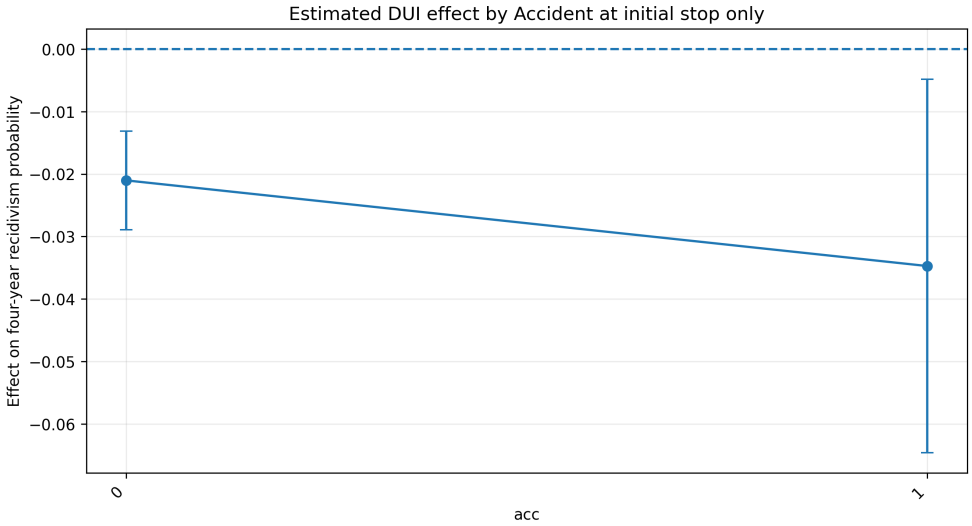


Figure 7: Estimated treatment-effect profile by accident involvement

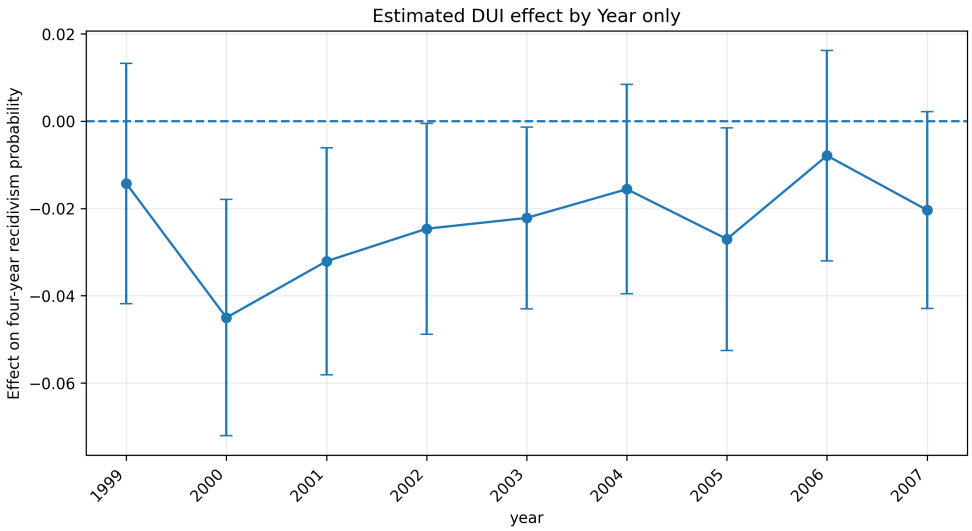


Figure 8: Estimated treatment-effect profile by year

A.5. Bandwidth sensitivity

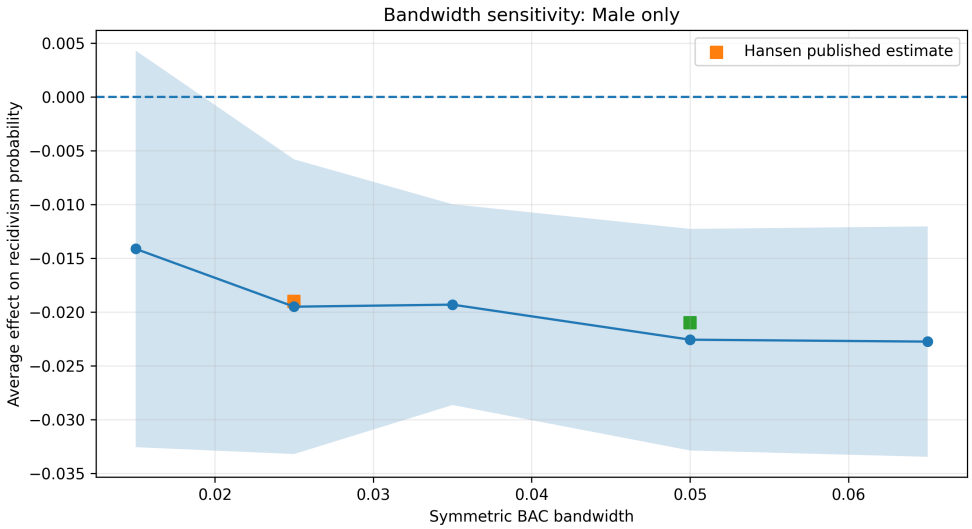


Figure 9: Gender-model sensitivity to bandwidth choice

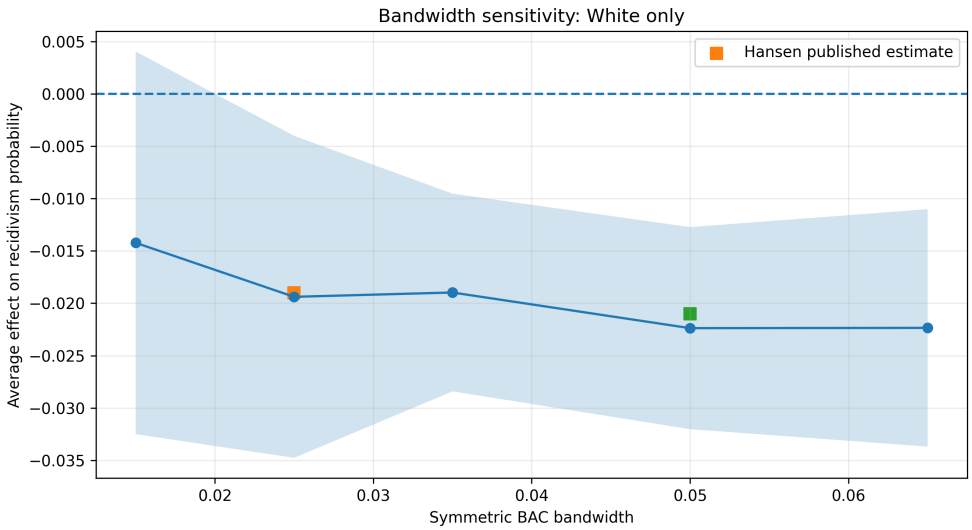


Figure 10: Race-model sensitivity to bandwidth choice

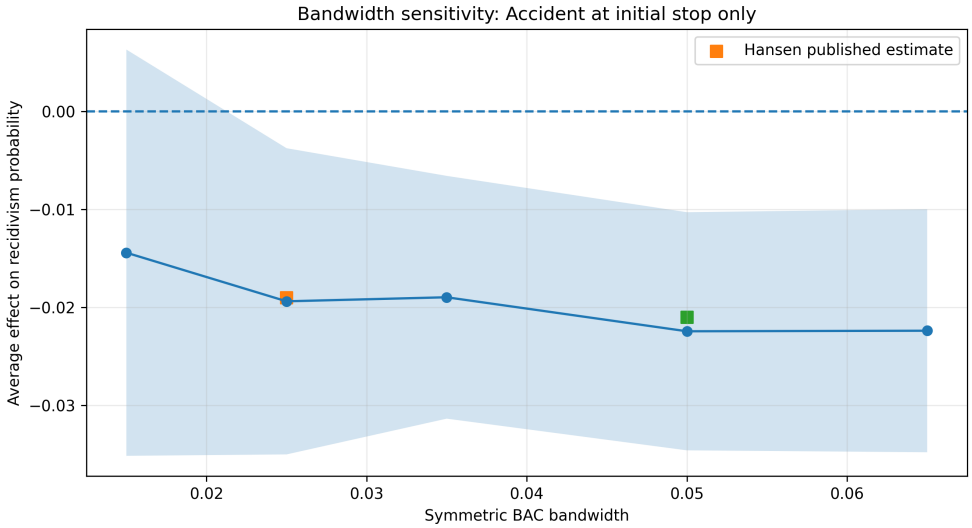


Figure 11: Accident-model sensitivity to bandwidth choice

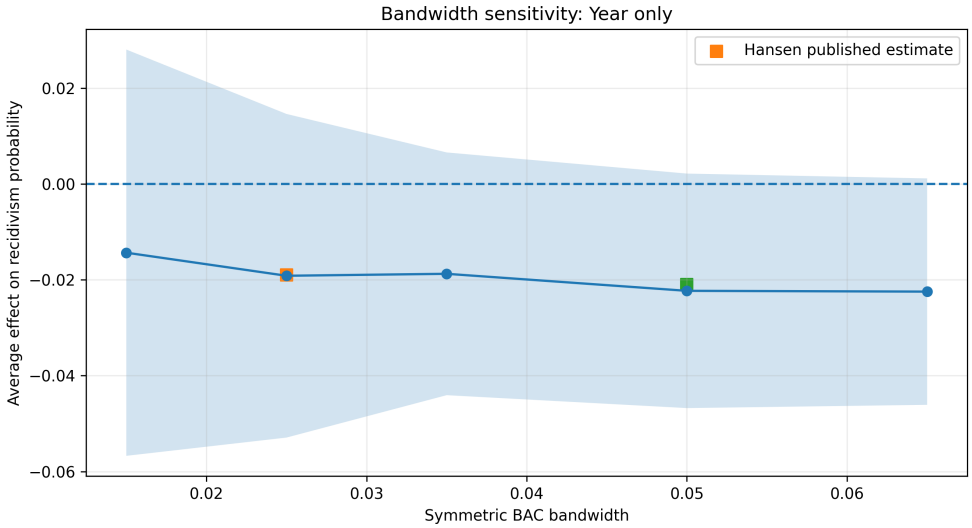


Figure 12: Year-model sensitivity to bandwidth choice